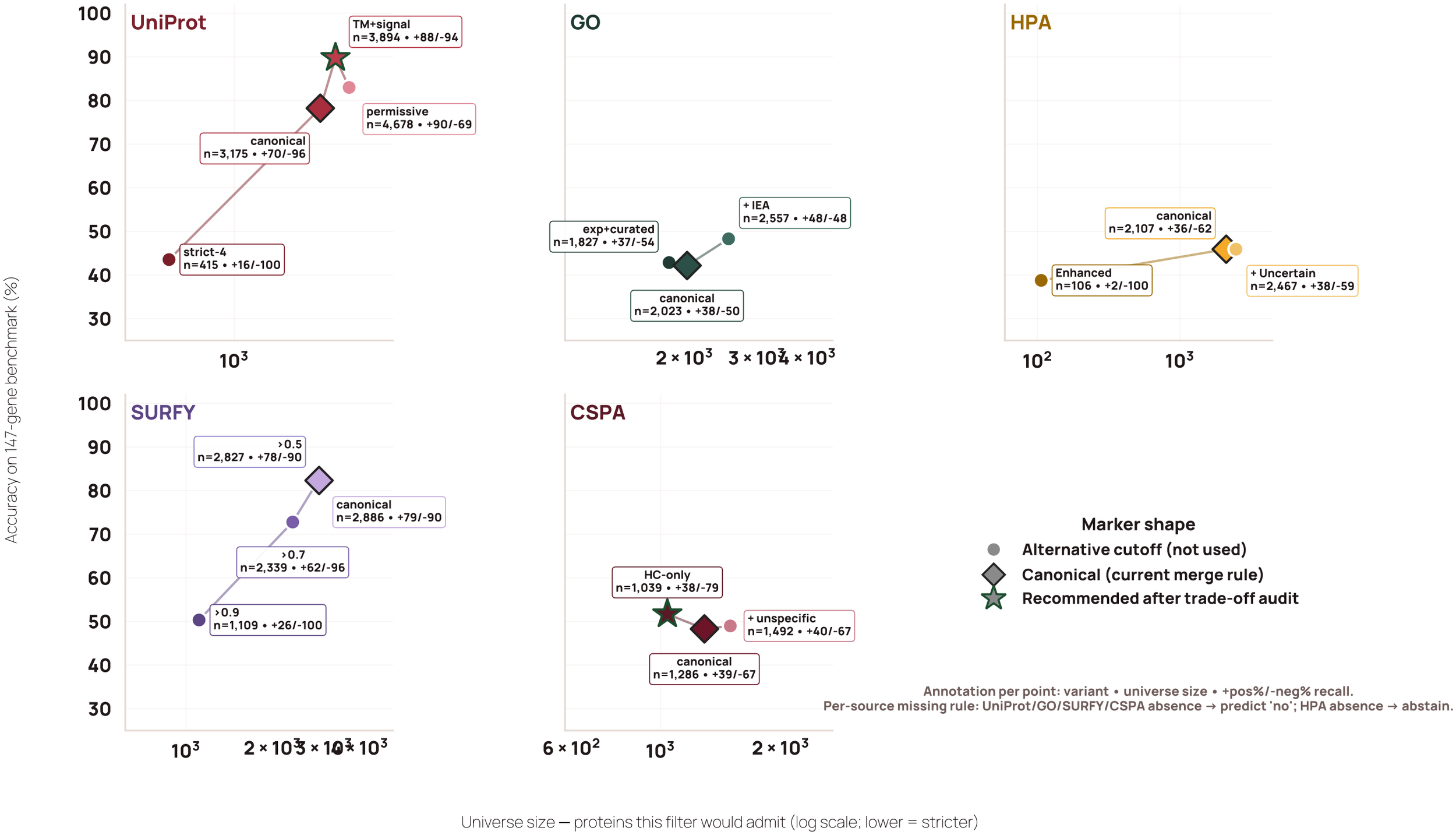
UniProt
TM+signal
n=3,894 • +88/-94
permissive
n=4,678 • +90/-69
canonical
n=3,175 • +70/-96
strict-4
n=415 • +16/-100
GO
+ IEA
n=2,557 • +48/-48
exp+curated
n=1,827 • +37/-54
canonical
n=2,023 • +38/-50
HPA
canonical
n=2,107 • +36/-62
Enhanced
n=106 • +2/-100
+ Uncertain
n=2,467 • +38/-59
SURFY
>0.5
n=2,827 • +78/-90
canonical
n=2,886 • +79/-90
>0.7
n=2,339 • +62/-96
>0.9
n=1,109 • +26/-100
CSPA
HC-only
n=1,039 • +38/-79
+ unspecific
n=1,492 • +40/-67
canonical
n=1,286 • +39/-67
Marker shape
Alternative cutoff (not used)
Canonical (current merge rule)
Recommended after trade-off audit
Annotation per point: variant • universe size • +pos%/-neg% recall.
Per-source missing rule: UniProt/GO/SURFY/CSPA absence → predict 'no'; HPA absence → abstain.
Accuracy on 147-gene benchmark (%)
Universe size — proteins this filter would admit (log scale; lower = stricter)